Lesson Planning

Complete Lesson Plan Book

Class: Class 8

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Linear Equations in One Variable

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Unit 1: Linear Equations in One Variable

Learning Objectives

• Students will recognise a linear equation in one variable and name its variable.

• Students will solve a linear equation by doing the same operation on both sides.

• Students will check a solution by substituting it back into the original equation.

Key Concepts

• A linear equation in one variable has one unknown, and the highest power of that unknown is 1.

• An equation is like a balanced scale: whatever is done to one side must be done to the other.

• Inverse operations undo each other, so addition undoes subtraction and multiplication undoes division.

• A solution is correct only if it makes the left side equal to the right side.

Teaching Methodology (Step by Step)

1. Start with a balance picture: if 3 plus a hidden number equals 10, what must the hidden number be?

2. Solve x plus 7 equals 15 together, showing the subtraction step on both sides.

3. Move to a two-step equation, 2x plus 3 equals 11, and undo the addition first, then the multiplication.

4. Stress the common mistake of doing a step on one side only, with a wrong example the class must correct.

5. Students solve four equations alone and check every answer by substitution.

Classroom Activities

• Equation relay: teams solve one step at a time, passing the board marker after each correct step.

• Find the error: three solved equations contain one mistake each, and students spot and fix them.

• Make your own: each student writes an equation with solution 6 and swaps it with a partner to solve.

Assessment Questions

1. Solve x minus 9 equals 4.

2. Solve 3x plus 5 equals 20, showing both steps.

3. A number is doubled and then 4 is added, giving 18. Write the equation and find the number.

Homework

• Solve six linear equations, checking each answer by substitution.

• Write a word problem that leads to the equation 4x minus 3 equals 17, and solve it.

Time Allocation

40 minutes